**SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE**

**Department of Artificial Intelligence and Machine Learning**

**Data Visualization Lab**

**Assignment Practical No. 3**

Power BI & Power Query

**Name:** Mahita Suresh Thundiyil

**PRN:** 24070126102

**Class:** AIML – B2

**Course:** Data Visualization Lab

**Course Code:** TE7663

**Academic Year:** 2026-2027

**Semester:** V

**Submitted To:** Dr. Sagar Pande

**Introduction**

This report documents the step-by-step execution of Experiment 3 for the Data Visualization Lab course. The Adventure Works dataset was loaded into Power BI Desktop, and the required built-in aggregations, calculated columns (using Power Query), and conditional/grouping tasks were performed on the appropriate worksheets.

Each task below includes the steps followed, the purpose of the transformation, and a screenshot of the result taken directly from Power BI.

**Tools Used**

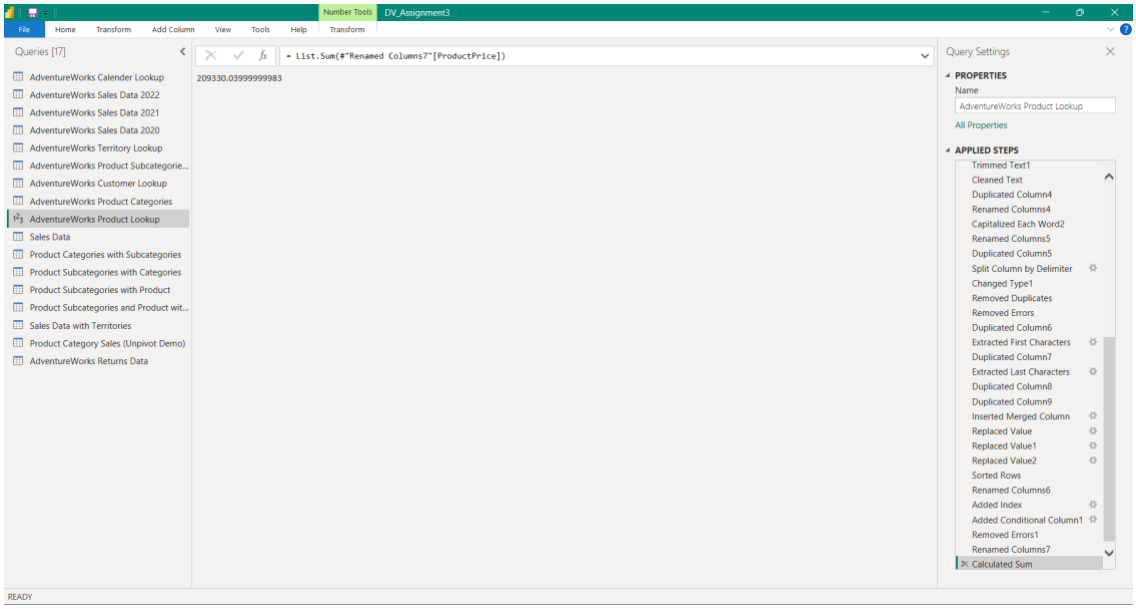
Microsoft Power BI Desktop — Power Query Editor for data transformation, and Card/Table visuals for built-in aggregations. Dataset used: Adventure Works.

**Part A — Built-in Aggregations**

Eight built-in aggregation functions were applied to key columns of the Adventure Works dataset using Card visuals in Power BI. Each aggregation and the steps followed are documented below.

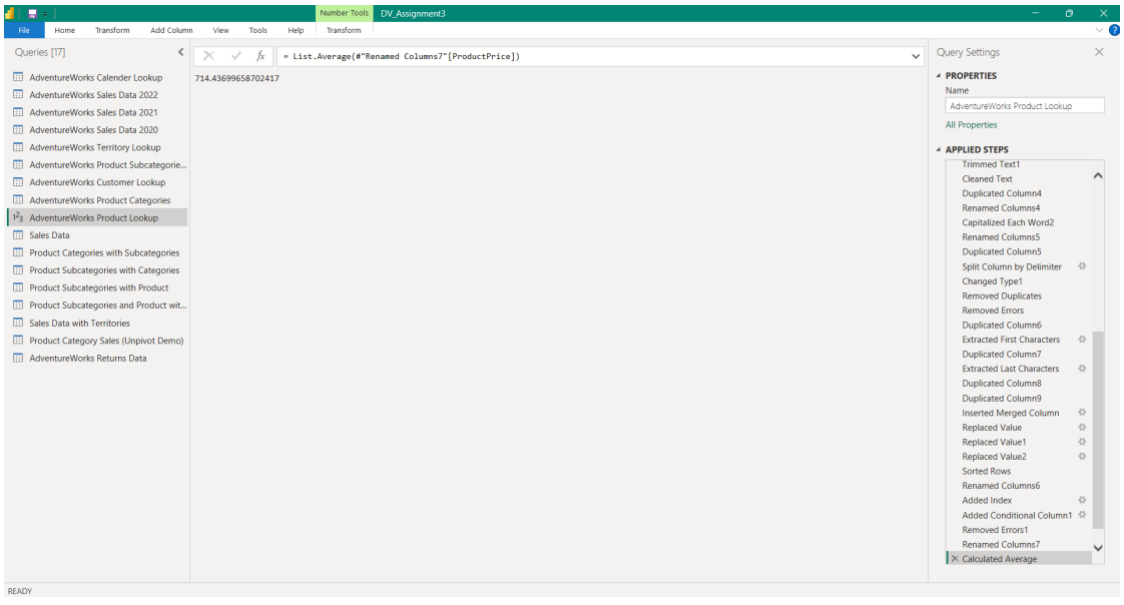
**1. SUM() on ProductPrice**

**Purpose:** Gives the total sum of all product prices in the Adventure Works dataset.



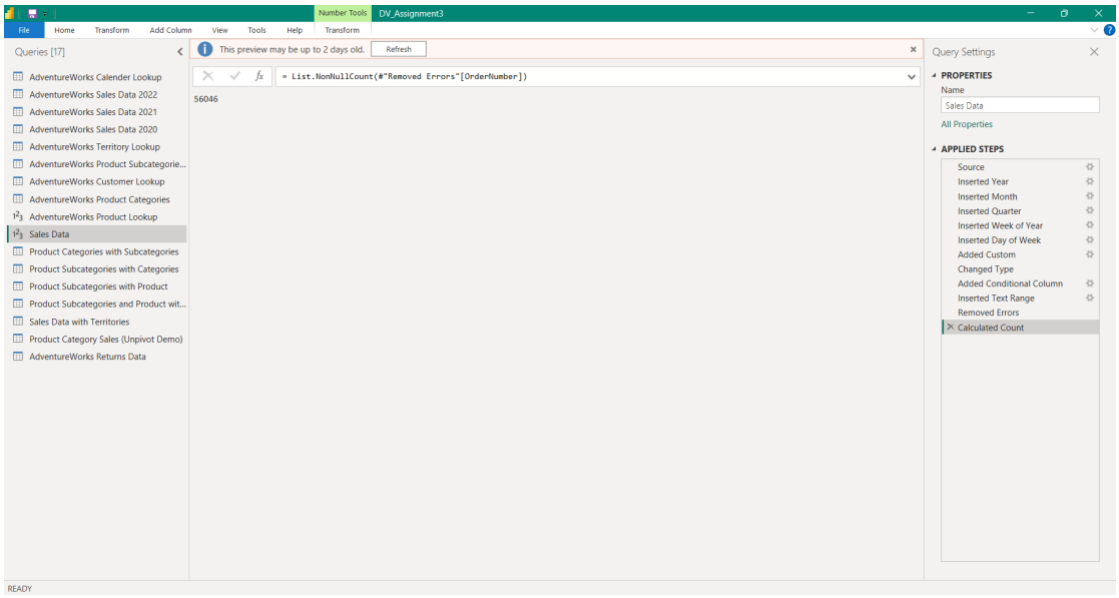

**2. AVERAGE() on ProductPrice**

**Purpose:** Shows the average price across all products, useful for pricing benchmarks.



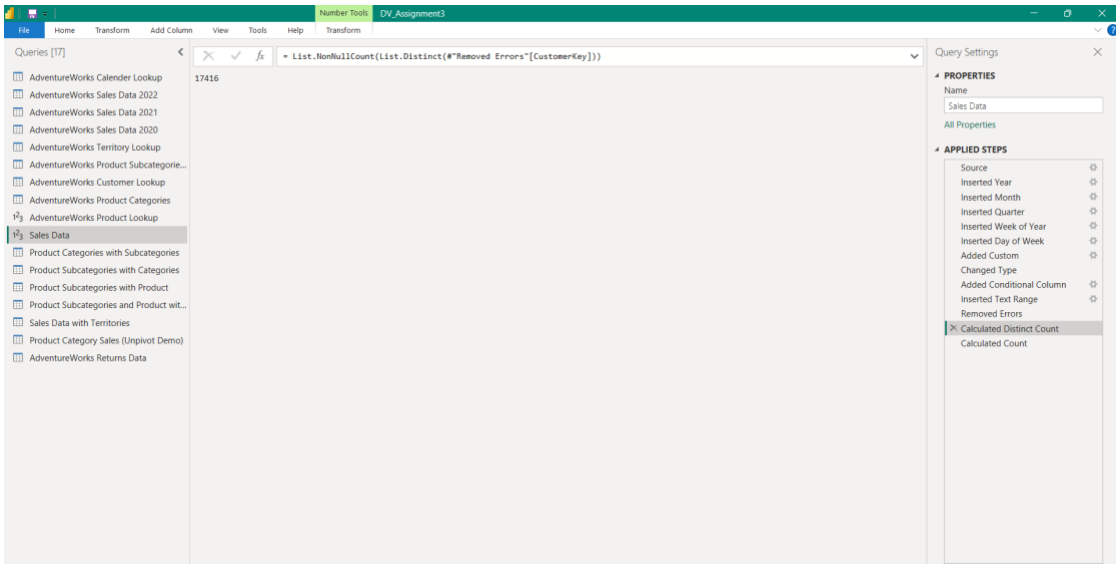
### 3. COUNT() on OrderNumber (Total Orders)

**Purpose:** Displays the total number of orders placed.



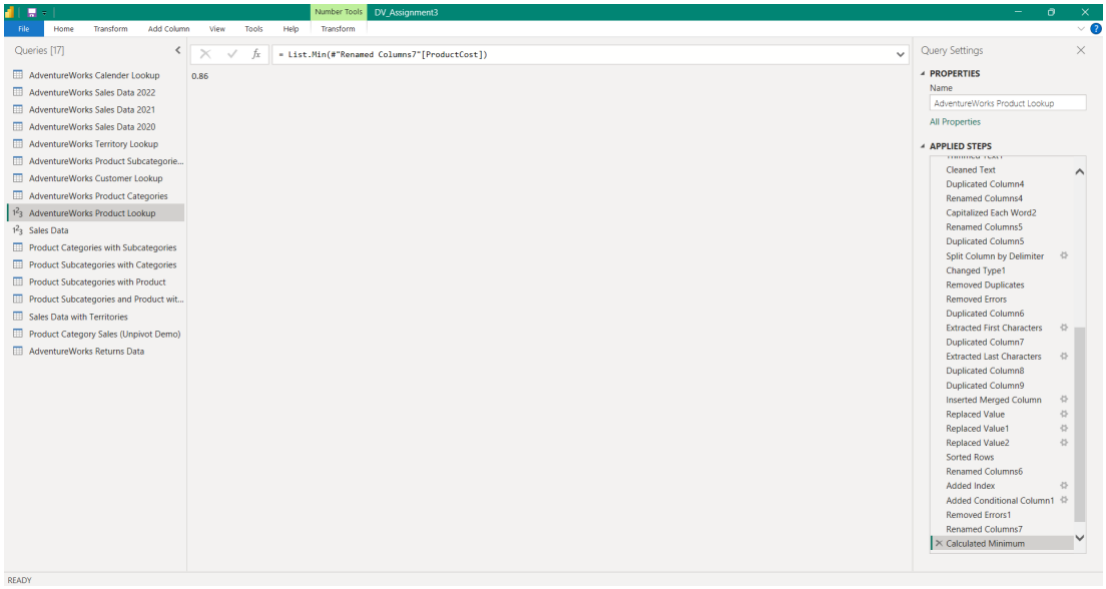

### 4. DISTINCTCOUNT() on CustomerKey (Total Customers)

**Purpose:** Shows the total number of unique customers in the dataset.



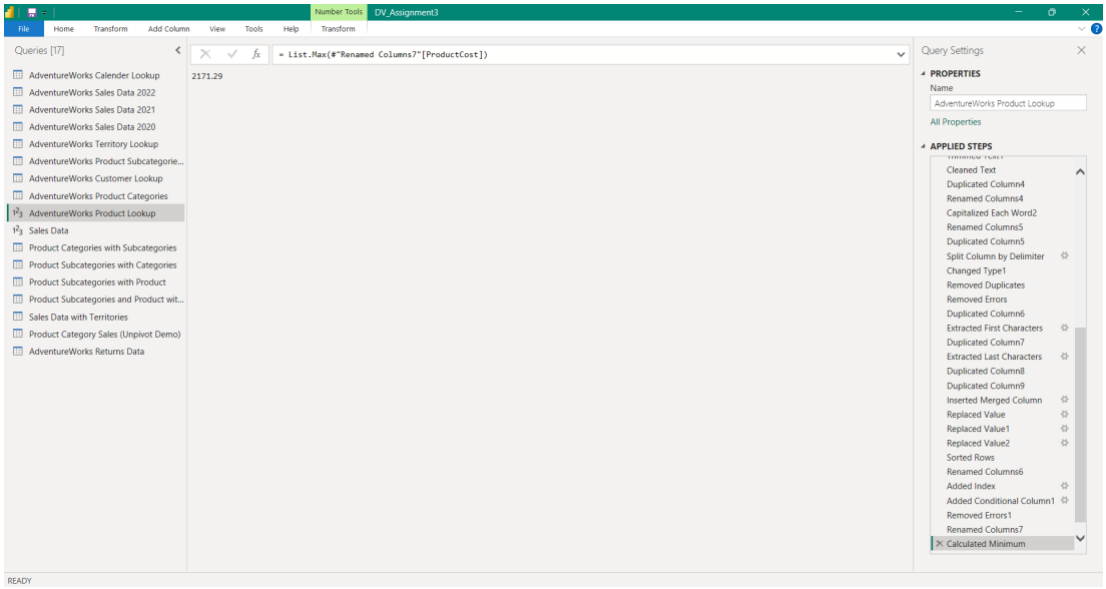

### 5. MIN() on ProductCost

**Purpose:** Displays the lowest product cost recorded in the dataset.



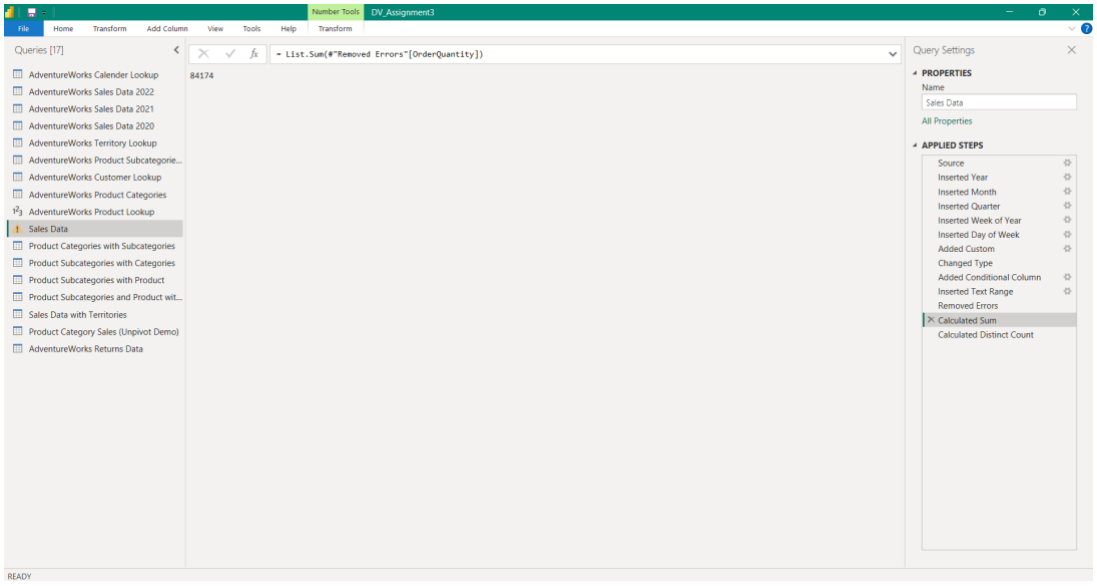

## 6. MAX() on ProductCost

**Purpose:** Displays the highest product cost recorded in the dataset.



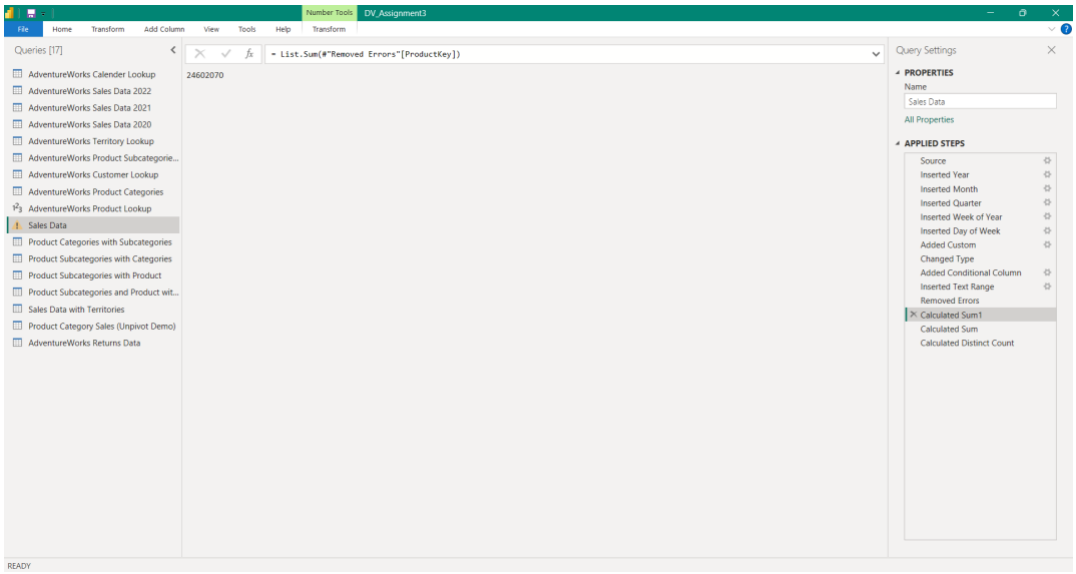

## 7. SUM() on OrderQuantity

**Purpose:** Gives the total quantity of products ordered across all transactions.



### 8. SUM() on Total Products

**Purpose:** Gives the overall count/sum of products available in the dataset.



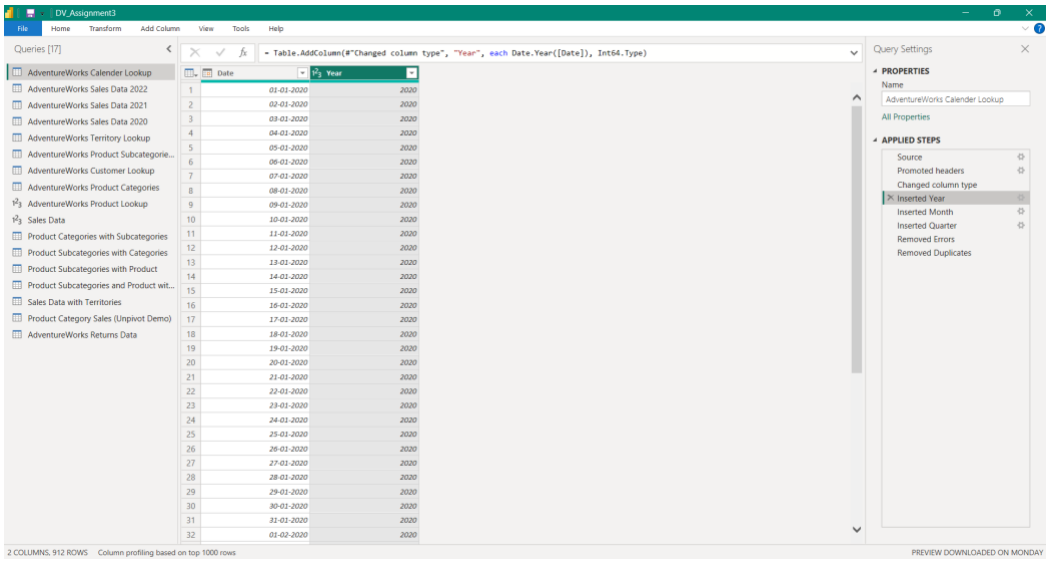

## Part B — Calculated Columns (Using Power Query)

The following 30 calculated columns were created on the Adventure Works dataset using Power Query's Add Column and Transform ribbons. Each task's steps and purpose are documented below along with a screenshot of the executed query.

### 1. Extract Order Year

**Steps:** Select OrderDate column → Add Column tab → Date → Year → Year.

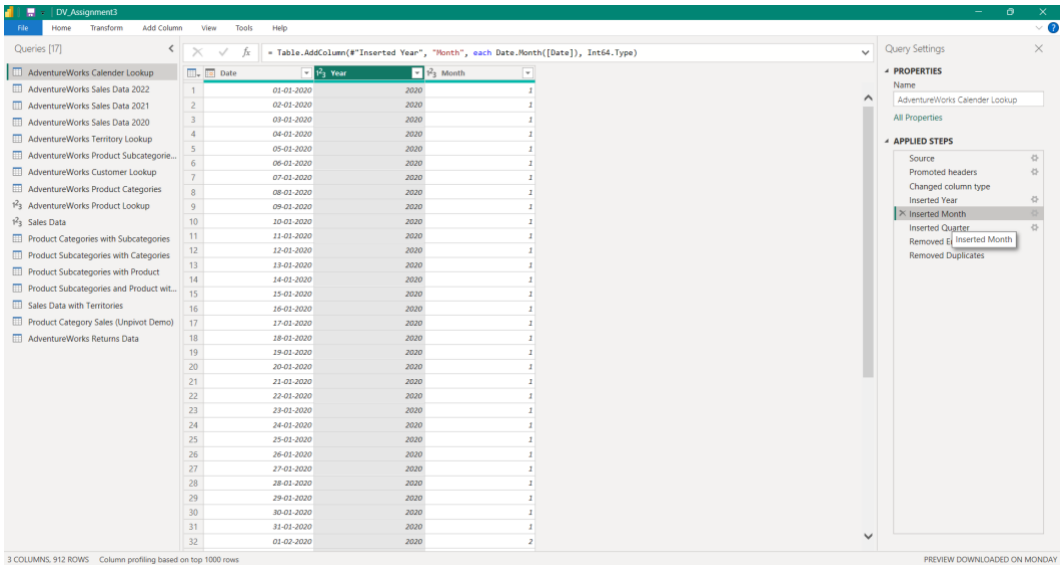
**Purpose:** Extracts the calendar year from the order date for year-wise trend analysis.



## 2. Extract Month Name

**Steps:** Select OrderDate column → Add Column → Date → Month → Name of Month.

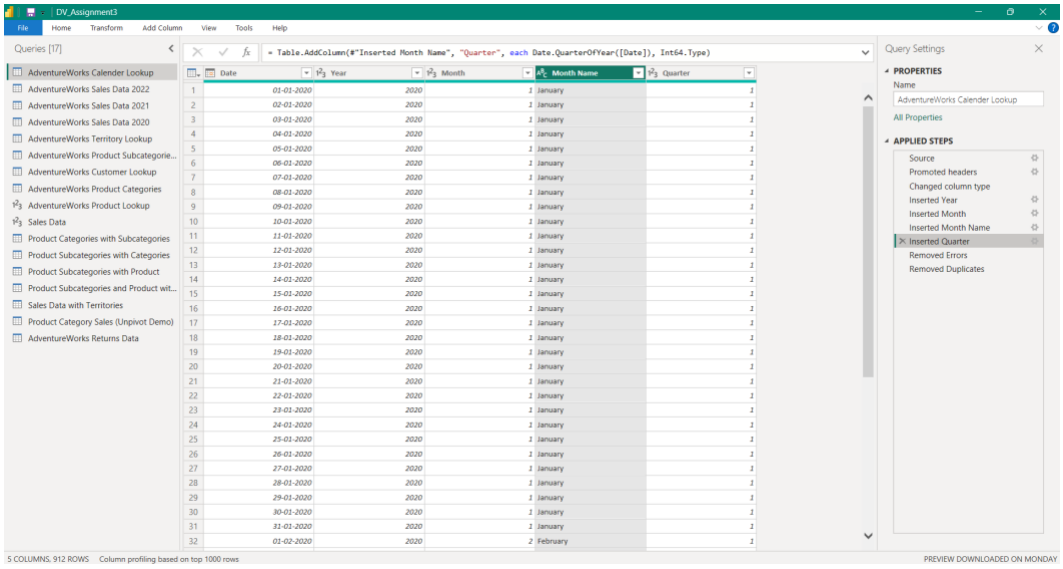
**Purpose:** Extracts the full month name from the order date for month-wise analysis.



## 3. Extract Quarter

**Steps:** Select OrderDate column → Add Column → Date → Quarter → Quarter of Year.

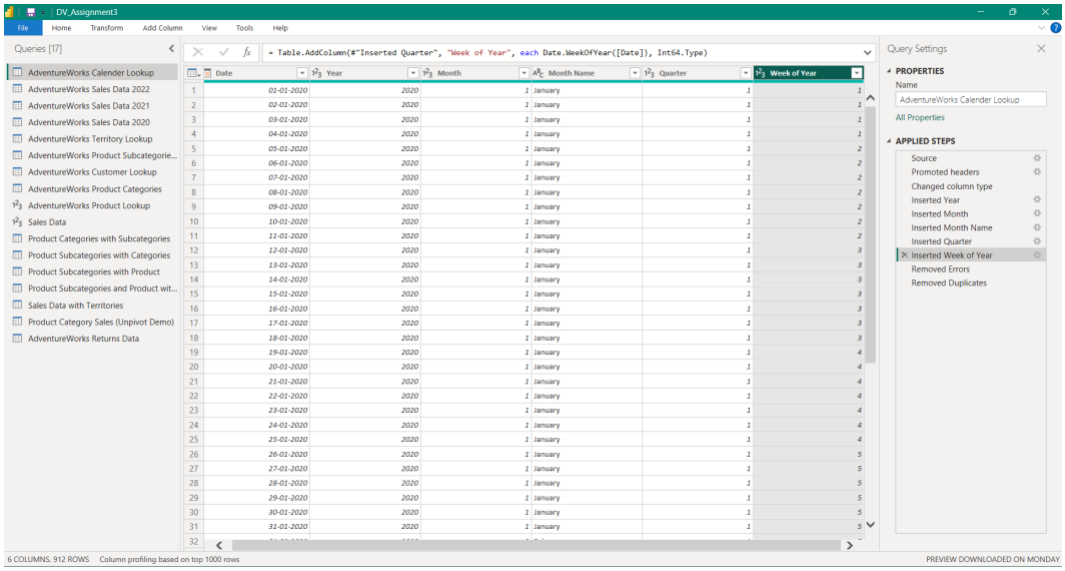
**Purpose:** Identifies which quarter (Q1–Q4) each order falls into.



## 4. Extract Week Number

**Steps:** Select OrderDate column → Add Column → Date → Week → Week of Year.

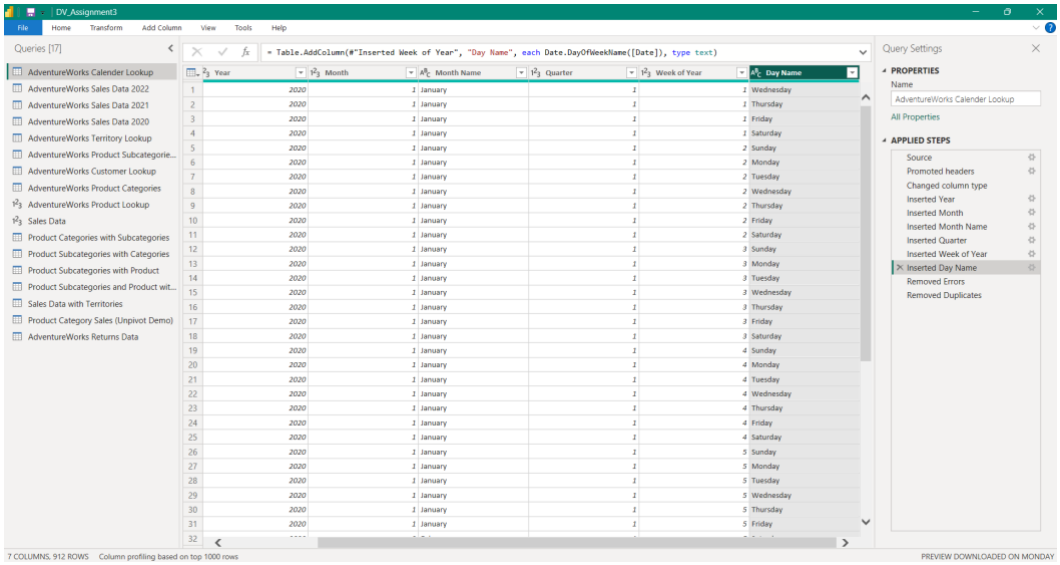
**Purpose:** Extracts the ISO week number of the order date, useful for weekly reporting.




## 5. Extract Day Name

**Steps:** Select OrderDate column → Add Column → Date → Day → Name of Day.

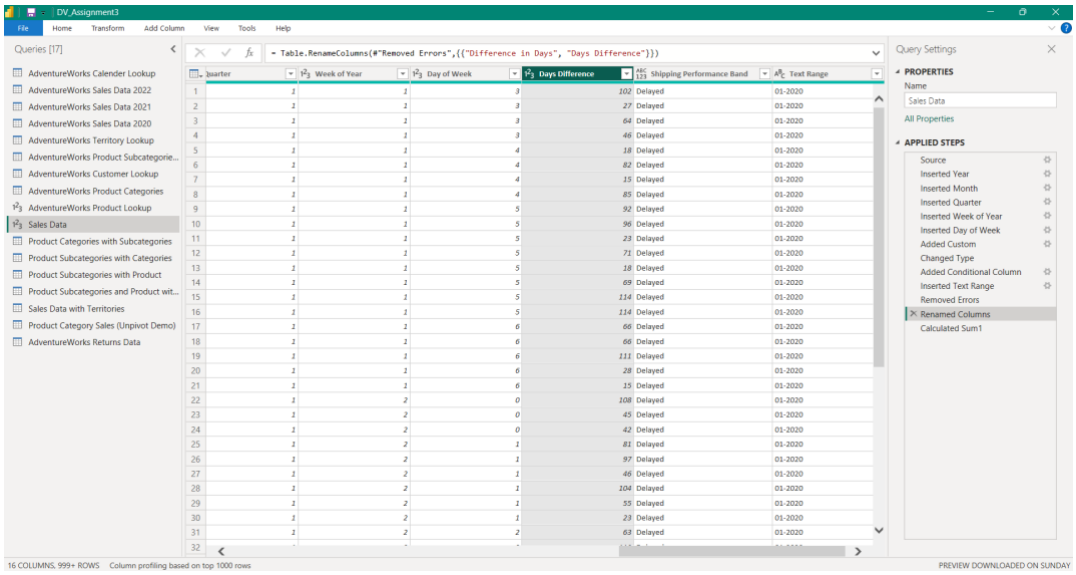
**Purpose:** Extracts the weekday name (Monday, Tuesday, etc.) from the order date.



## 6. Calculate Days Difference

**Steps:** Select OrderDate then Ctrl-select StockDate → Add Column → Date → Subtract Days.

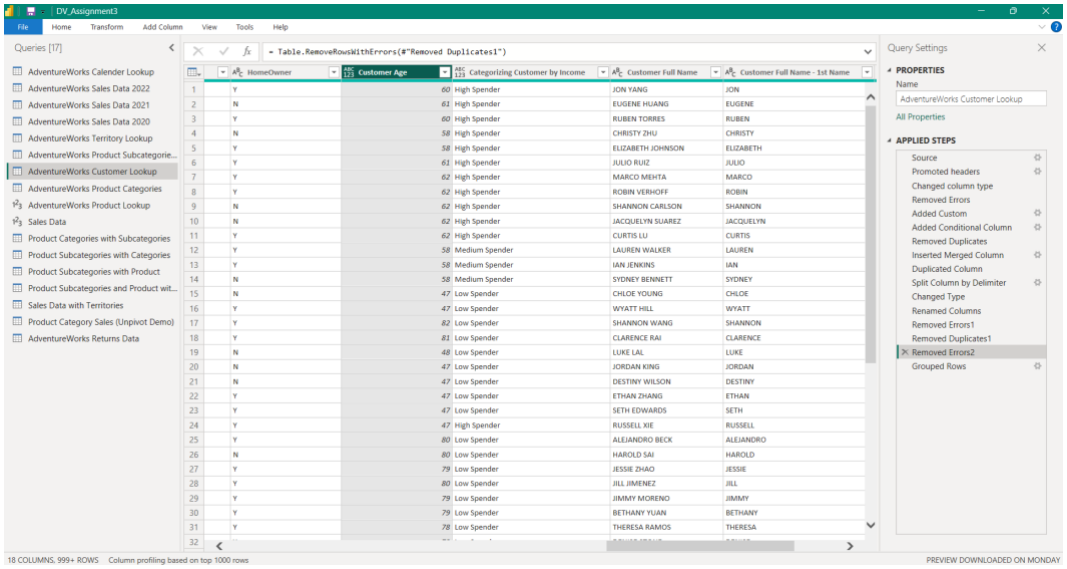
**Purpose:** Calculates the number of days between the stock date and the order date.



## 7. Calculate Customer Age

**Steps:** Select BirthDate column → Add Column → Date → Age.

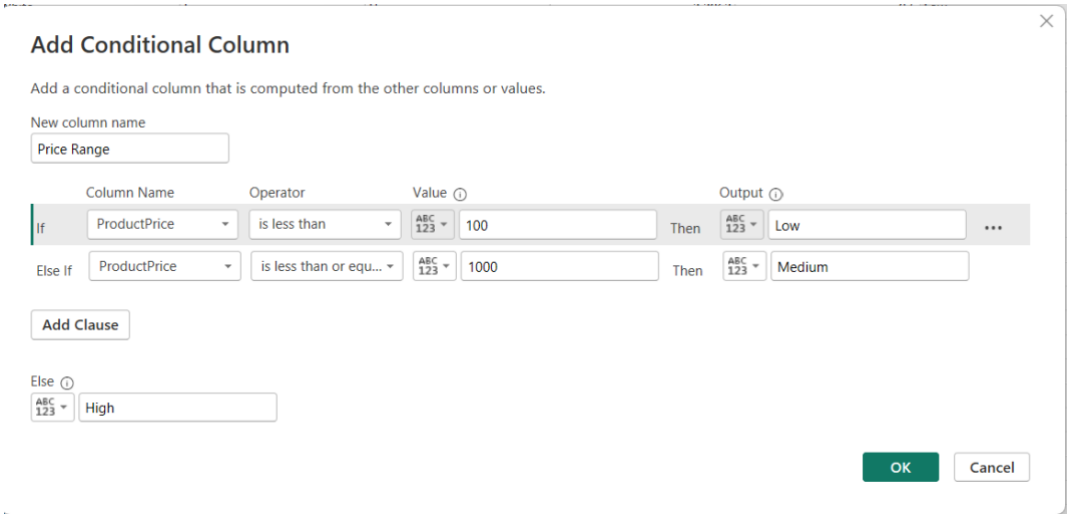
**Purpose:** Calculates each customer's current age from their date of birth.

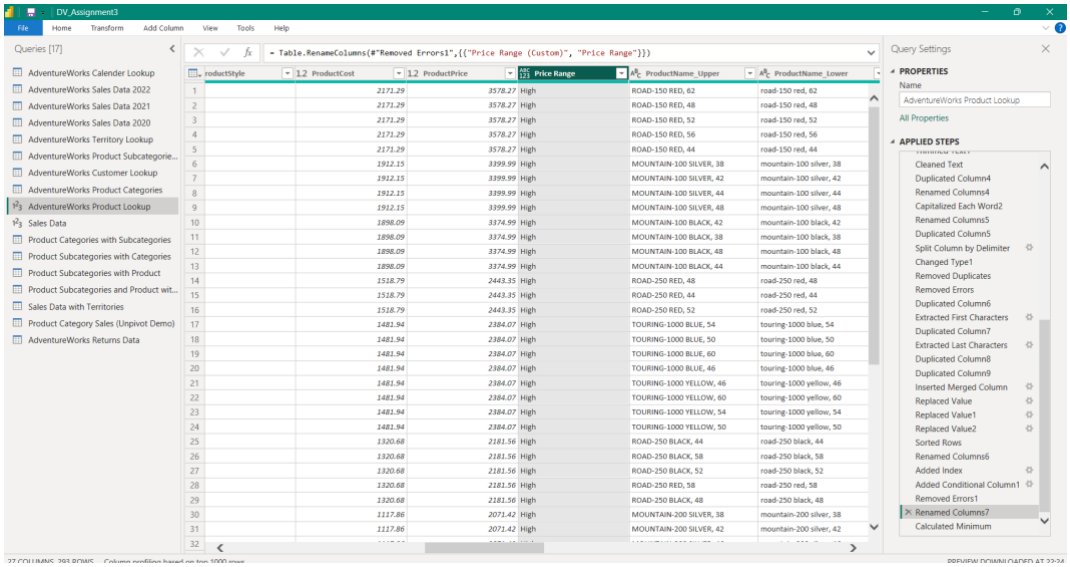


## 8. Create Price Range (Low/Medium/High)

**Steps:** Add Column → Conditional Column. If ProductPrice ≤ 100 then "Low", else if ≤ 1000 then "Medium", else "High".

**Purpose:** Buckets products into price bands for easier comparison.

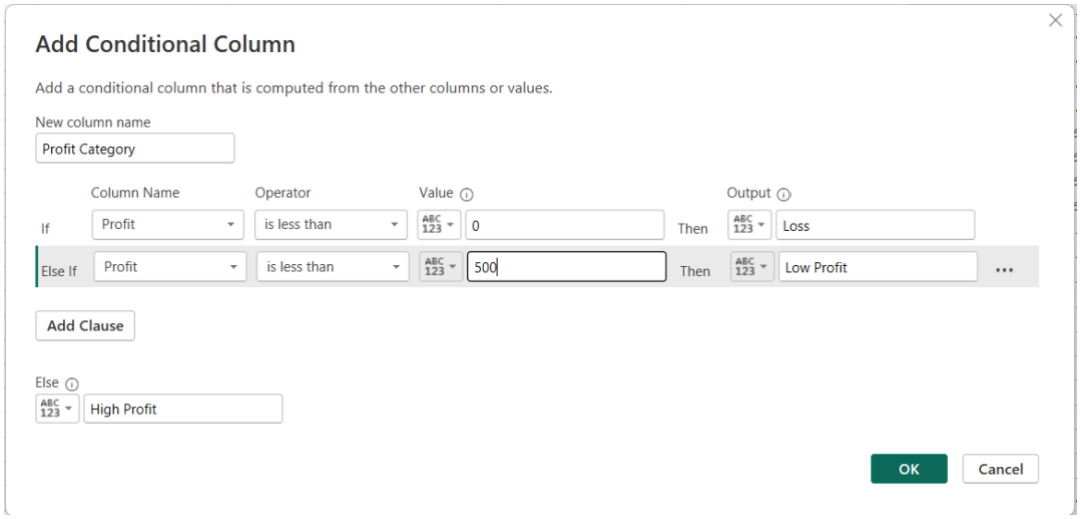


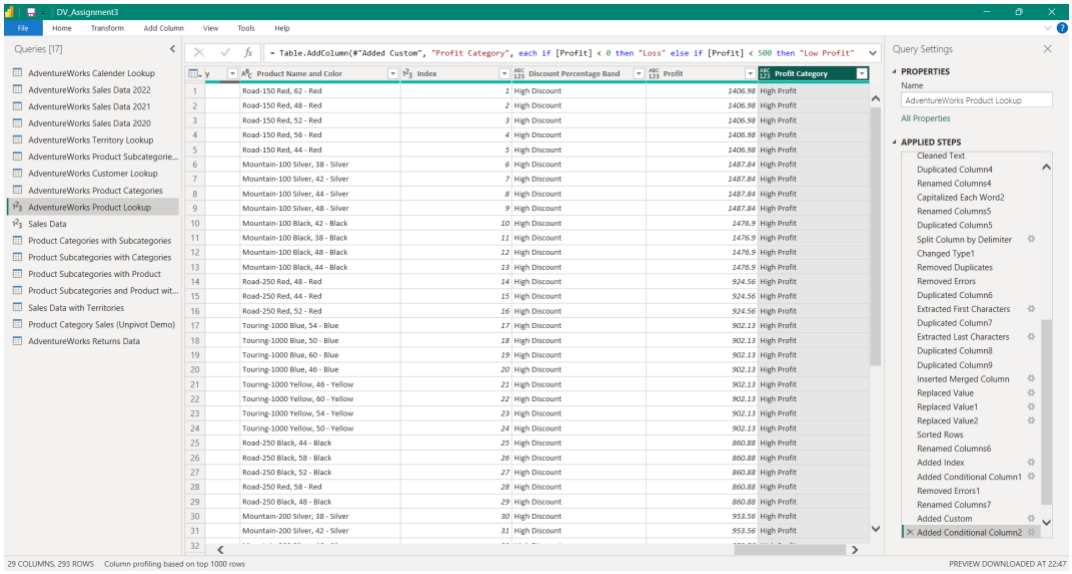



## 9. Create Profit Category (Loss/Low/High Profit)

**Steps:** Add Column → Conditional Column based on (ProductPrice − ProductCost). Negative → "Loss", small positive → "Low Profit", large positive → "High Profit".

**Purpose:** Classifies each product by its profitability.

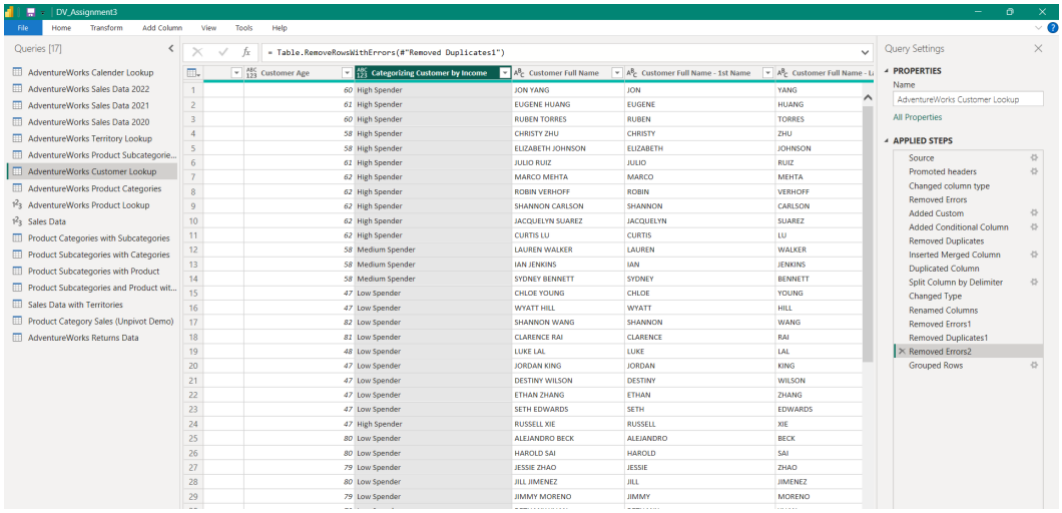


## 10. Categorize Customers based on Income

**Steps:** Add Column → Conditional Column based on a customer's income (e.g. High / Medium / Low spender).

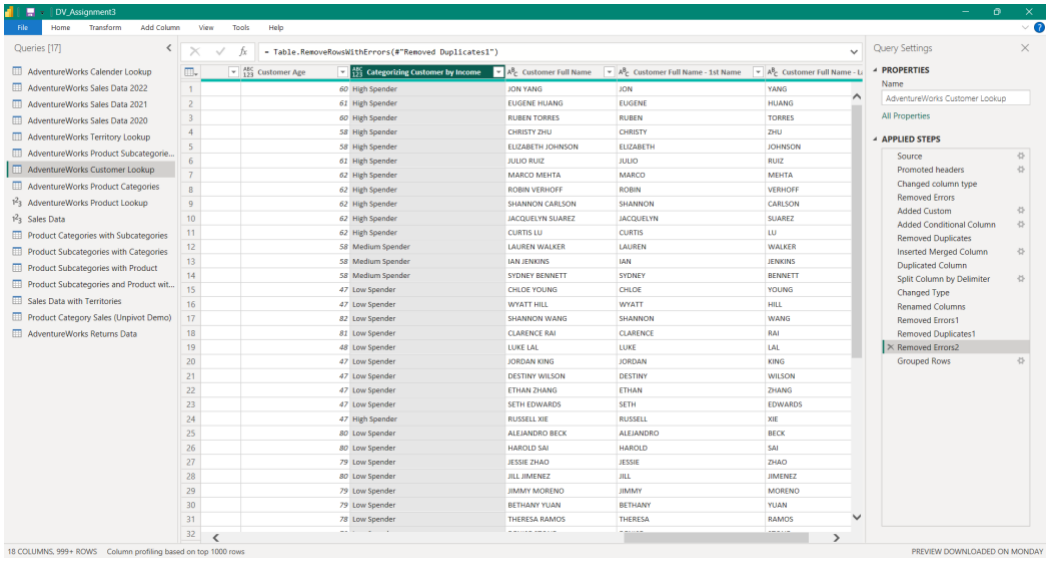
**Purpose:** Segments customers according to how much they have spent.



## 11. Round Price Amount

**Steps:** Select ProductPrice column → Transform tab → Rounding → Round.

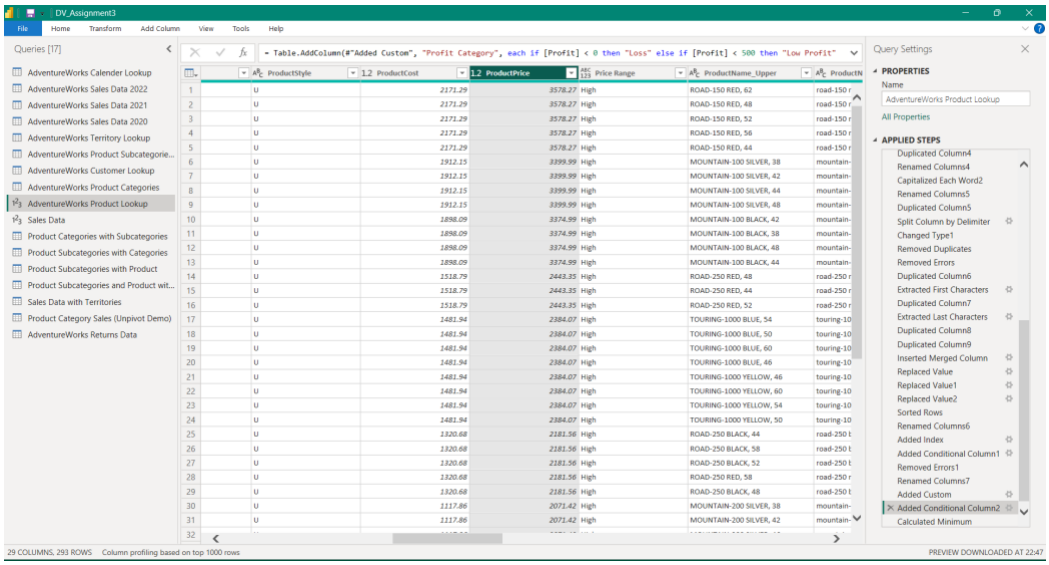
**Purpose:** Rounds the price value to a whole number for cleaner reporting.




## 12. Round Cost Amount (2 decimals)

**Steps:** Select ProductCost column → Transform → Rounding → Round → 2 decimal places.

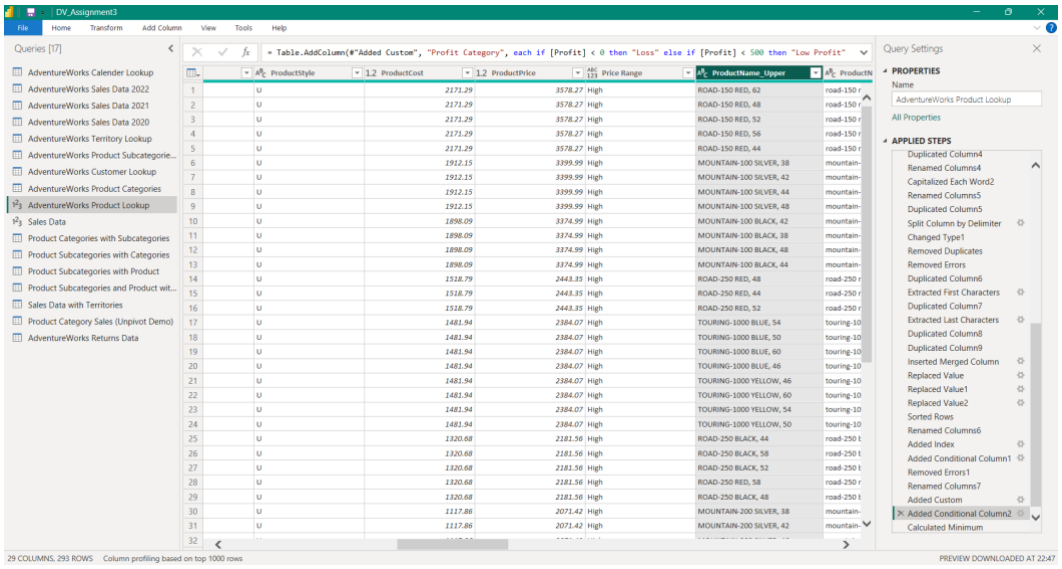
**Purpose:** Rounds the cost value to 2 decimal places for currency-style formatting.



## 13. Convert Product Name to UPPERCASE

**Steps:** Select Product Name column → Transform → Format → UPPERCASE.

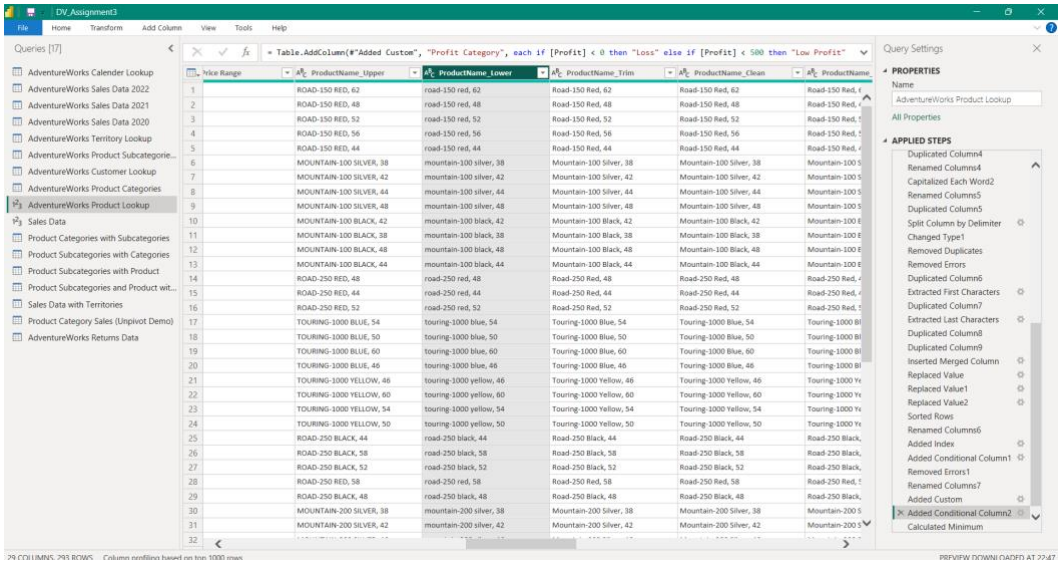
**Purpose:** Standardizes product names to all capital letters.



## 14. Convert Product Name to lowercase

**Steps:** Select Product Name column → Transform → Format → lowercase.

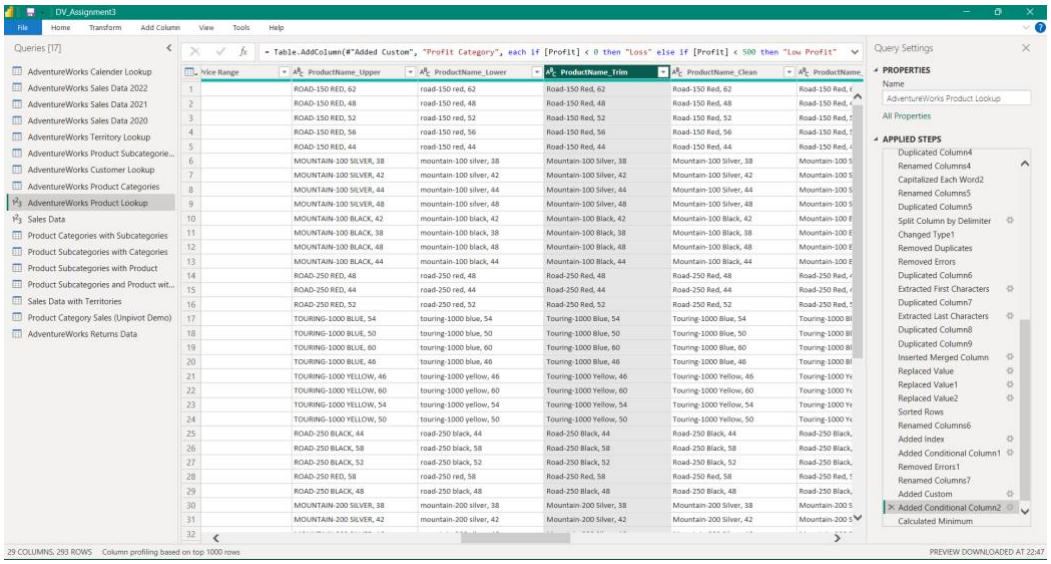
**Purpose:** Standardizes product names to all lowercase letters.



## 15. Trim Extra Spaces

**Steps:** Select Product Name column → Transform → Format → Trim.

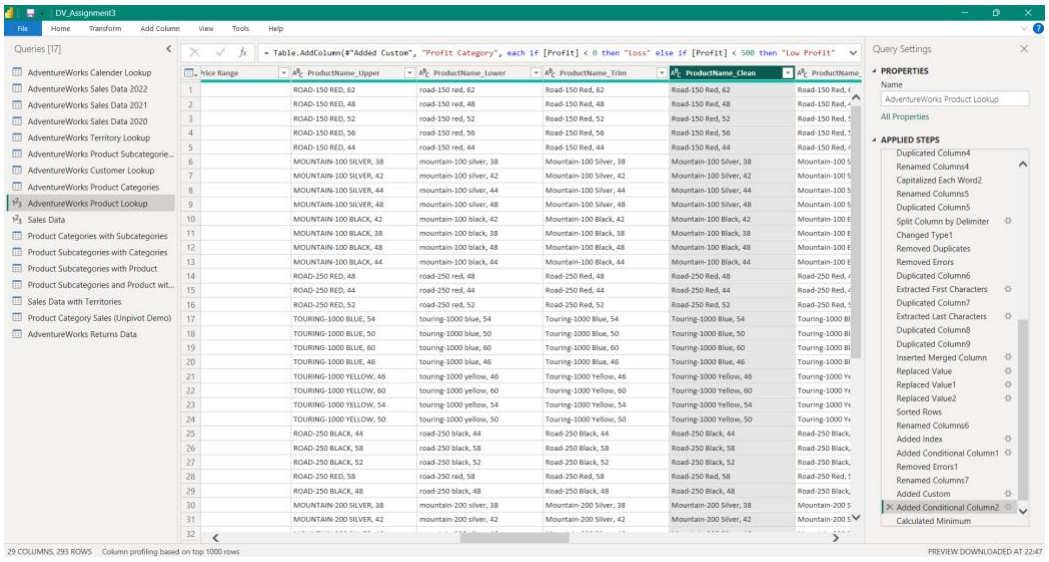
**Purpose:** Removes unwanted leading/trailing spaces from text values.



## 16. Clean Product Name

**Steps:** Select Product Name column → Transform → Format → Clean.

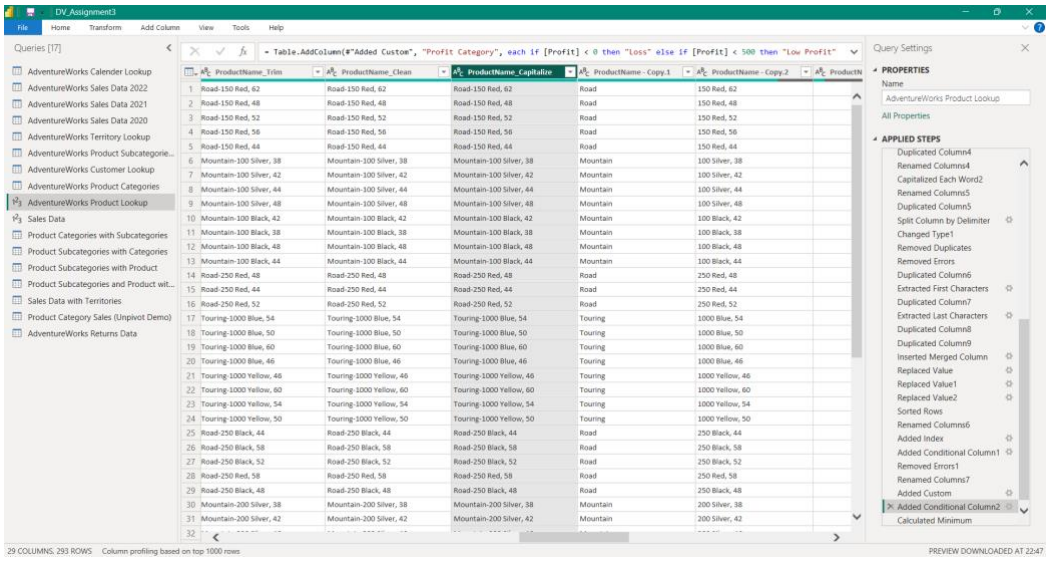
**Purpose:** Removes non-printable characters from the product name text.



## 17. Capitalize Each Word

**Steps:** Select Product Name column → Transform → Format → Capitalize Each Word.

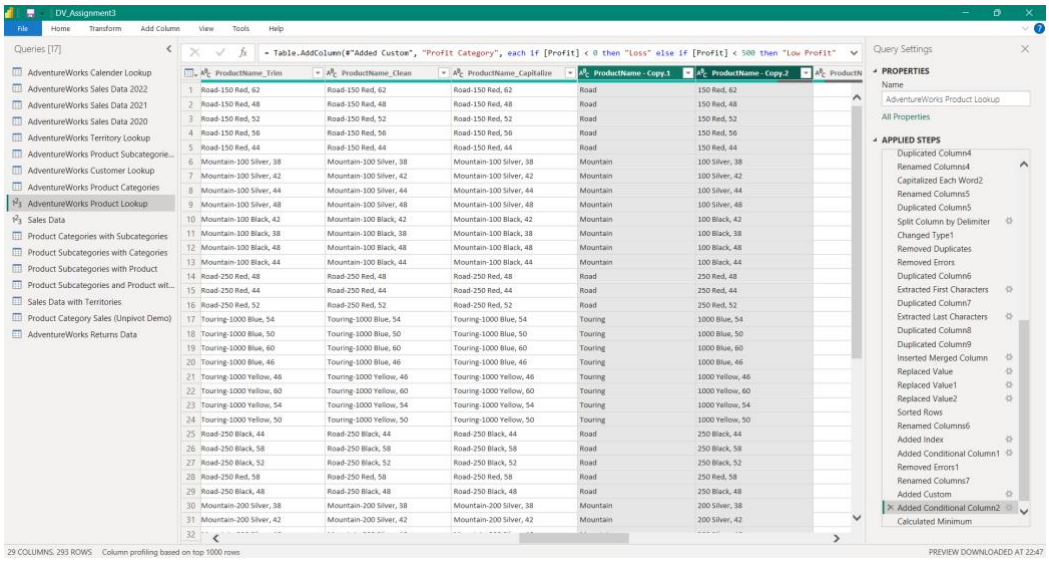
**Purpose:** Capitalizes the first letter of every word in the product name.



## 18. Split Product Name by "-"

**Steps:** Select Product Name column → Split Column → By Delimiter → "-".

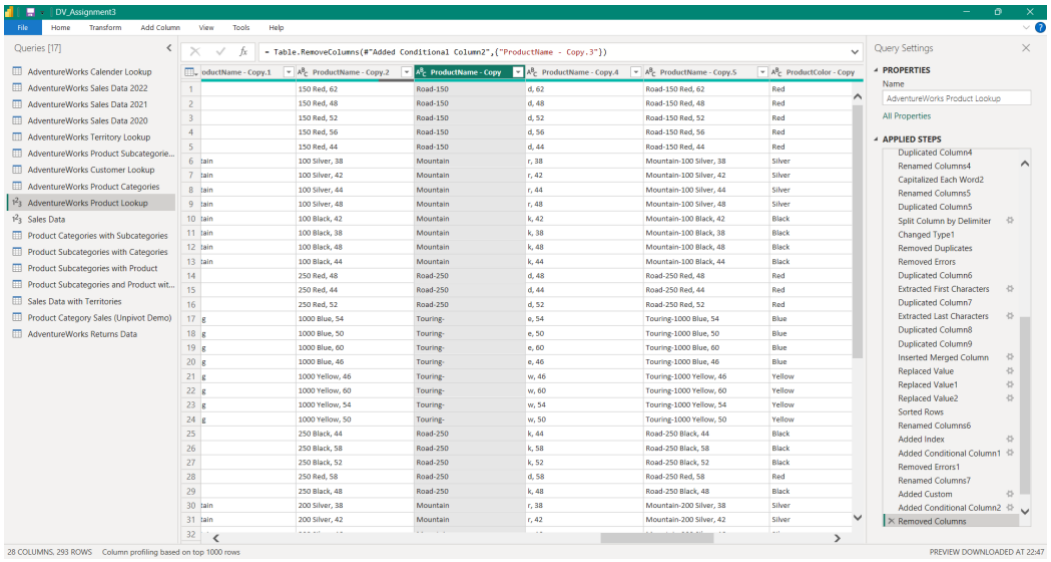
**Purpose:** Splits the product name into separate columns wherever a hyphen occurs.



## 19. Extract First 8 Characters

**Steps:** Select Product Name column → Add Column → Extract → First Characters → 8.

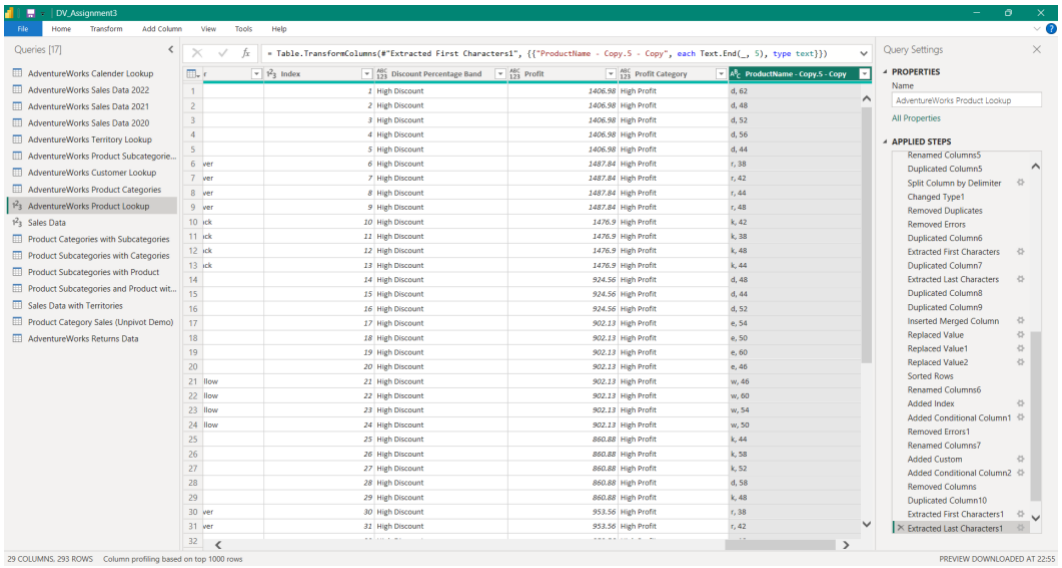
**Purpose:** Extracts the first 8 characters of the product name into a new column.



## 20. Extract Last 5 Characters

**Steps:** Select Product Name column → Add Column → Extract → Last Characters → 5.

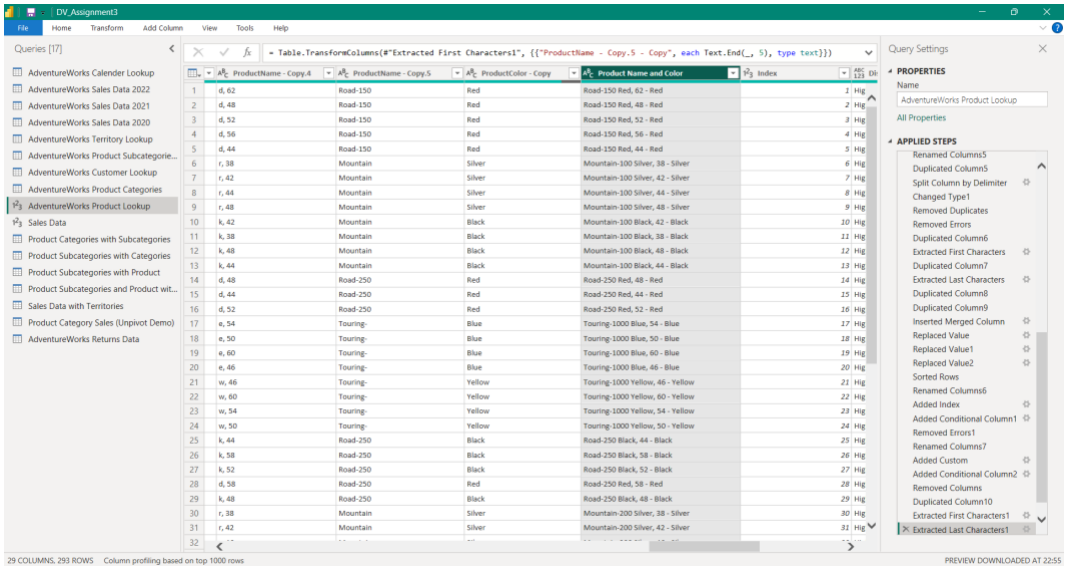
**Purpose:** Extracts the last 5 characters of the product name into a new column.



## 21. Merge Product Name and Color

**Steps:** Ctrl-select Product Name and Color columns → Transform (or Add Column) → Merge Columns, with a space/comma separator.

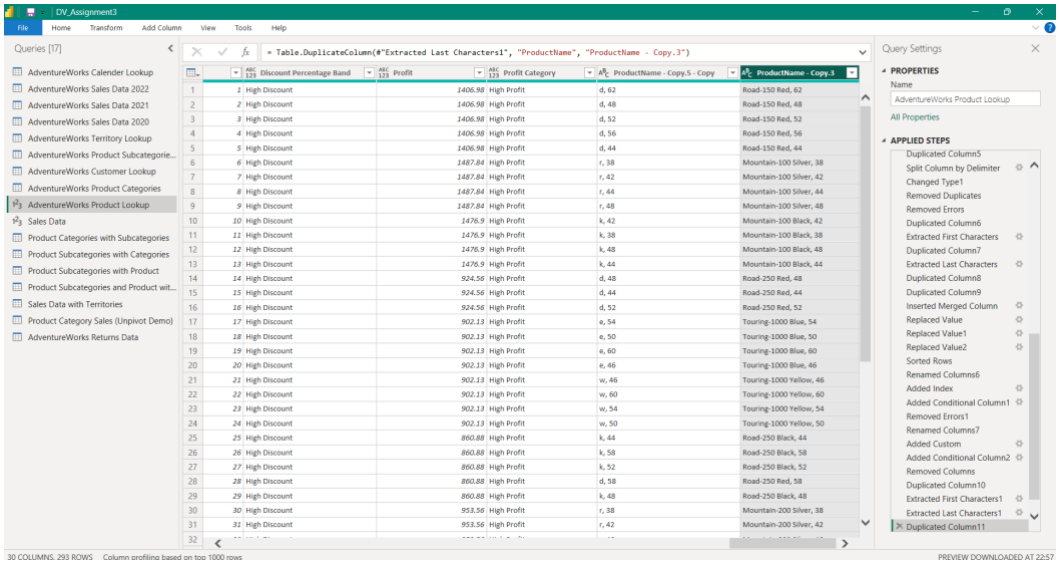
**Purpose:** Combines product name and color into one descriptive text column.



## 22. Duplicate Product Name Column

**Steps:** Right-click Product Name column → Duplicate Column.

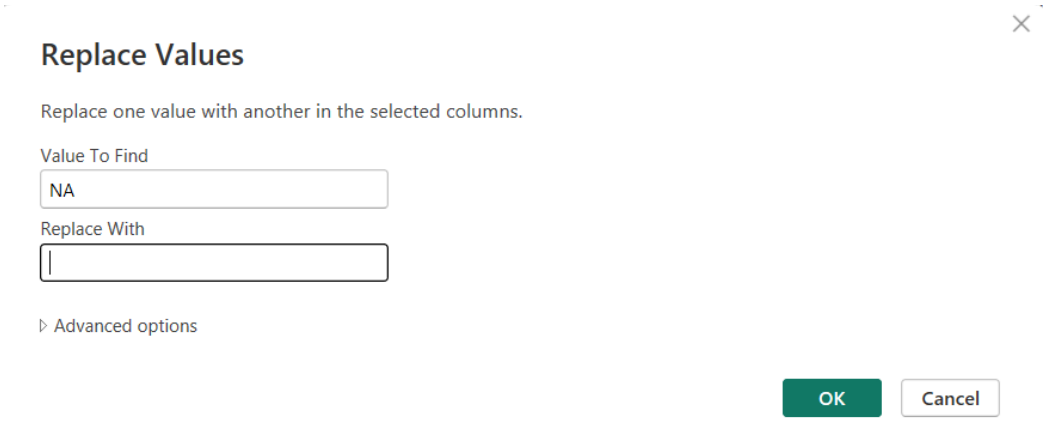
**Purpose:** Creates a copy of the Product Name column for further independent transformation.



### 23. Replace Missing Values

**Steps:** Right-click the relevant column → Replace Values → replace null with a default value.

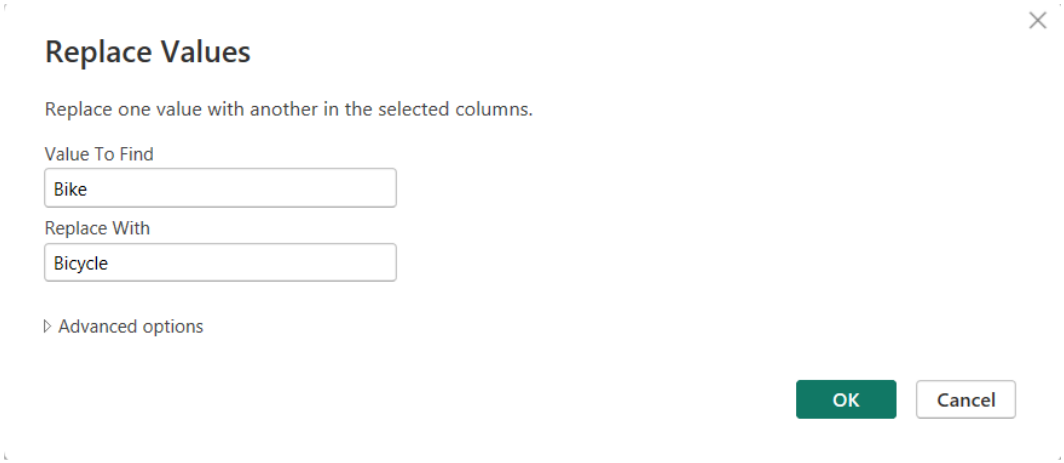
**Purpose:** Fills blank/null cells with a suitable default so downstream calculations don't break.




### 24. Replace Bike with Bicycle

**Steps:** Right-click Product Name column → Replace Values → "Bike" → "Bicycle".

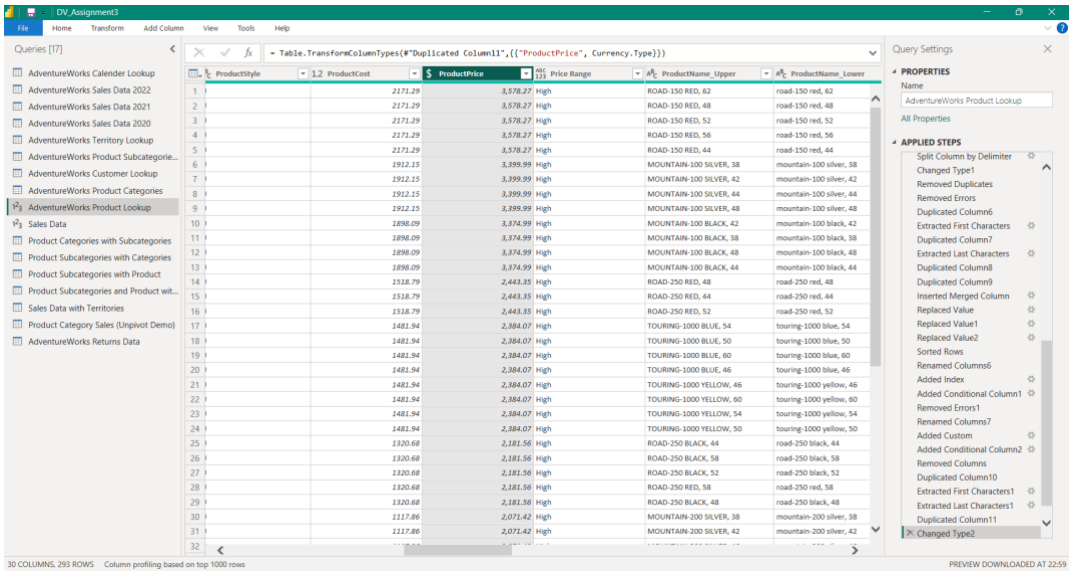
**Purpose:** Standardizes terminology by replacing the word "Bike" with "Bicycle".




### 25. Change Sales Data Type to Fixed Decimal

**Steps:** Click the data-type icon in the Sales column header → Fixed Decimal Number.

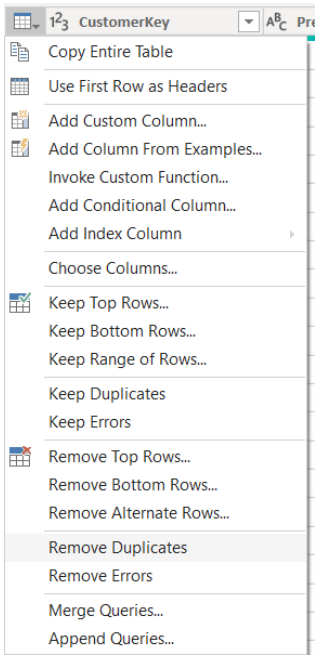
**Purpose:** Ensures the Sales column is stored with consistent decimal precision.

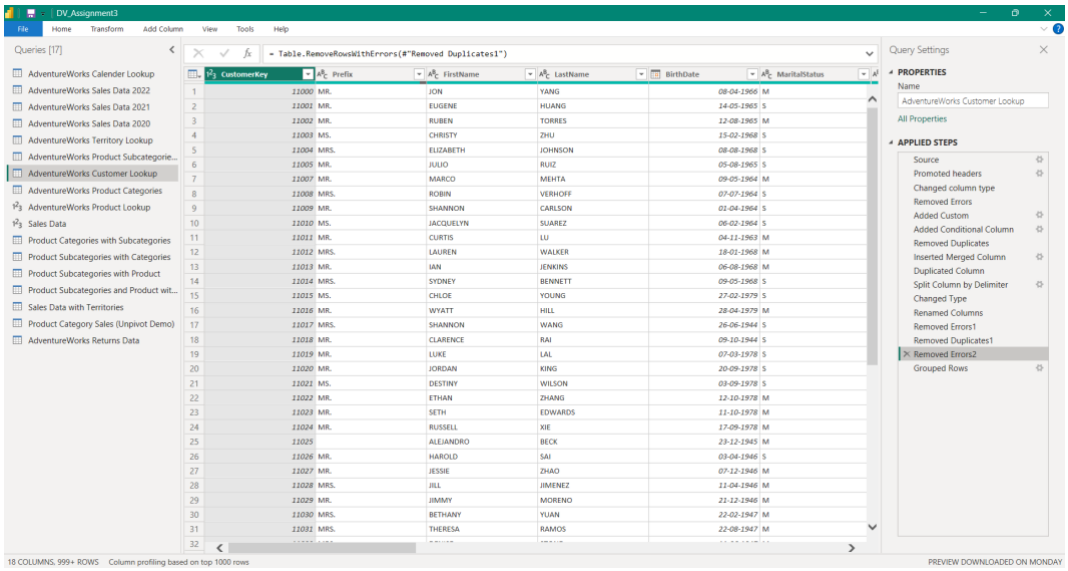


## 26. Remove Duplicate Customers

**Steps:** Right-click CustomerKey column → Remove Duplicates.

**Purpose:** Removes repeated customer records so each customer appears only once.

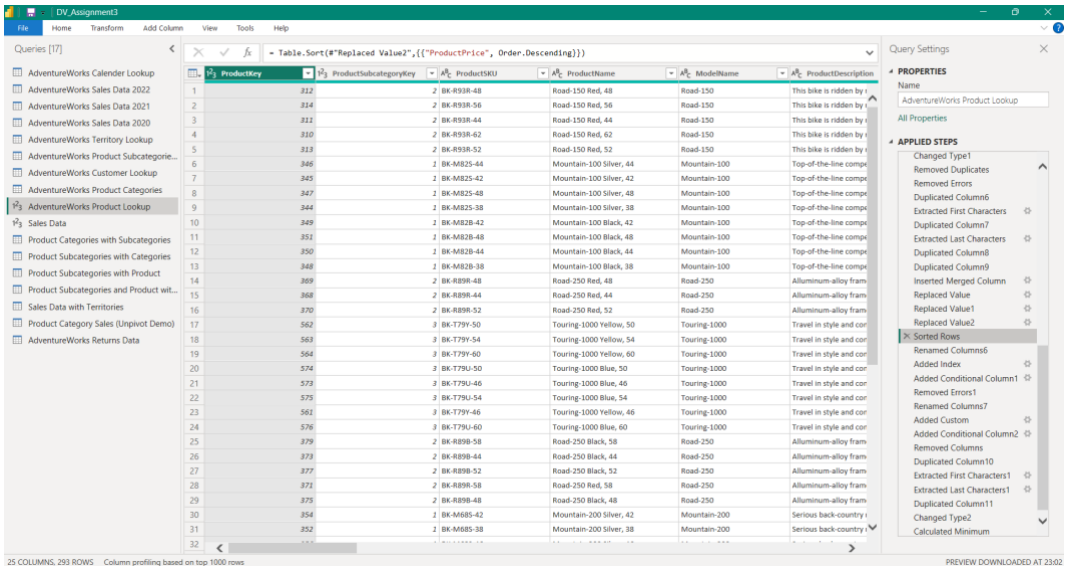


## 27. Sort Products by Sales Amount

**Steps:** Click the dropdown on the Sales column header → Sort Ascending/Descending.

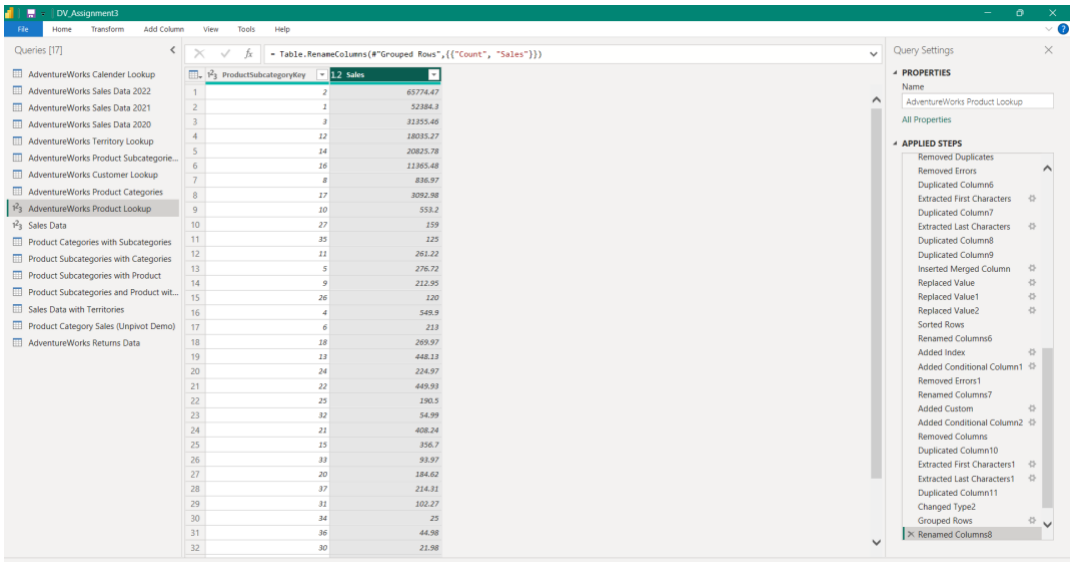
**Purpose:** Orders the product rows by their sales amount.



## 28. Group by Product Category and Sum Sales

**Steps:** Select ProductCategory column → Transform → Group By → New column name → Operation: Sum → Column: Sales.

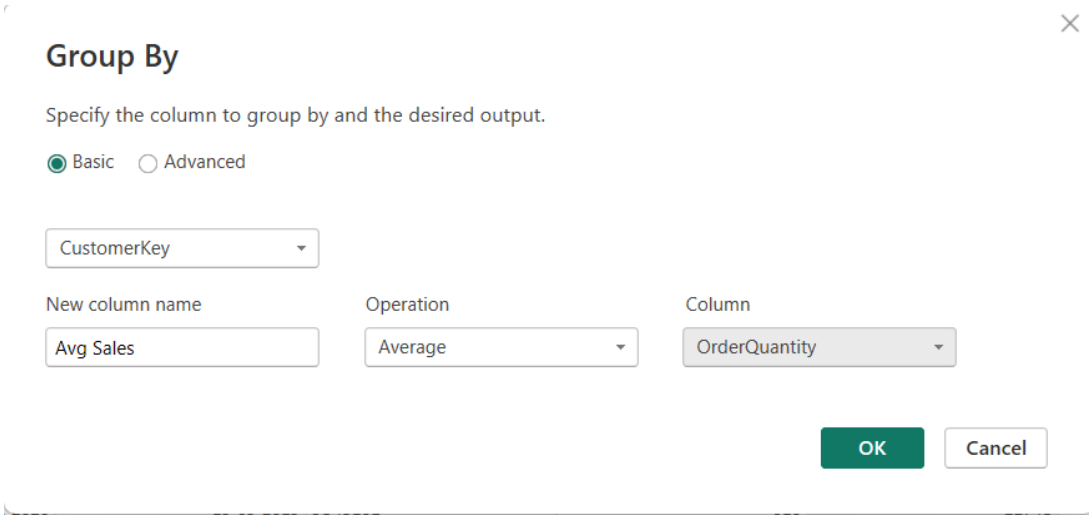
**Purpose:** Aggregates total sales for each product category.

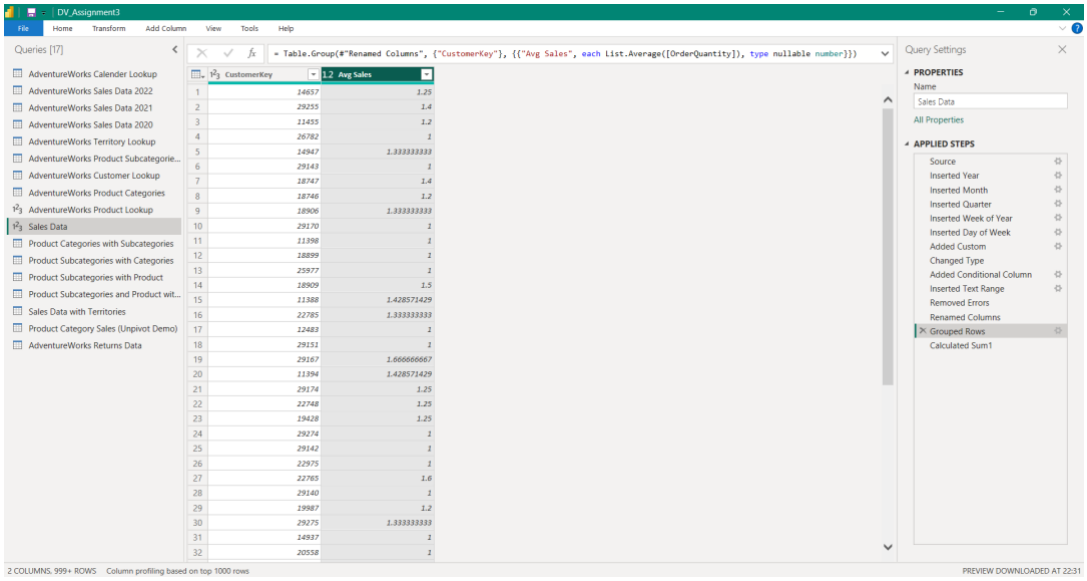


### 29. Group by Customer and Average Sales

**Steps:** Select CustomerKey column → Transform → Group By → Operation: Average → Column: Sales.

**Purpose:** Calculates the average sales value for each customer.

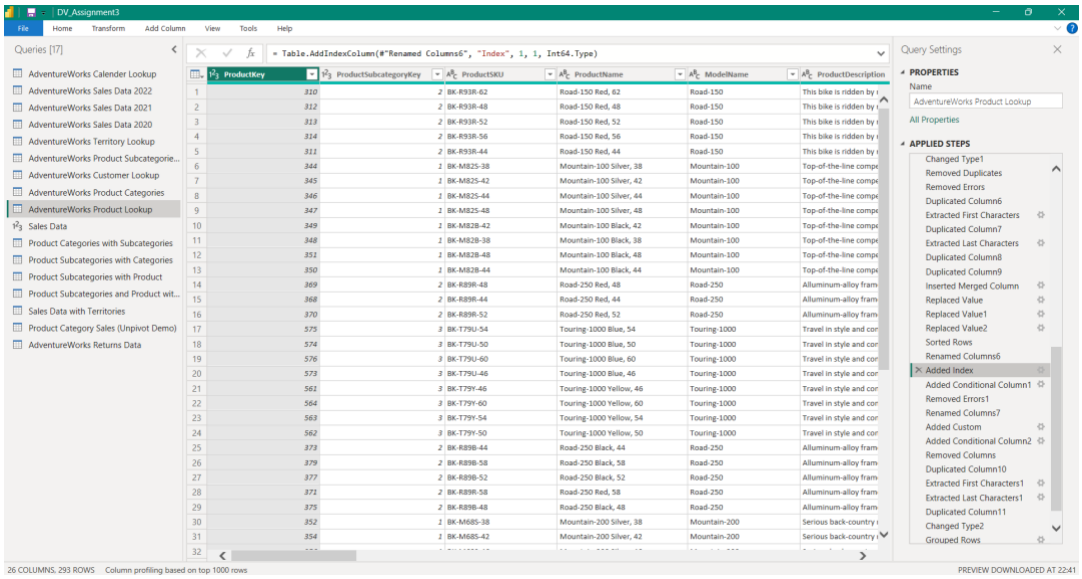


### 30. Create Index Column

**Steps:** Add Column tab → Index Column → From 0 (or From 1).

**Purpose:** Adds a sequential row number column, useful as a unique row identifier.




## Part C — Additional Conditional & Grouping Tasks

In addition to the tasks above, the following conditional columns, text/date transformations, and grouped aggregations were created to enrich the dataset for analysis.

### 1. Create Sales Band (Conditional Column)

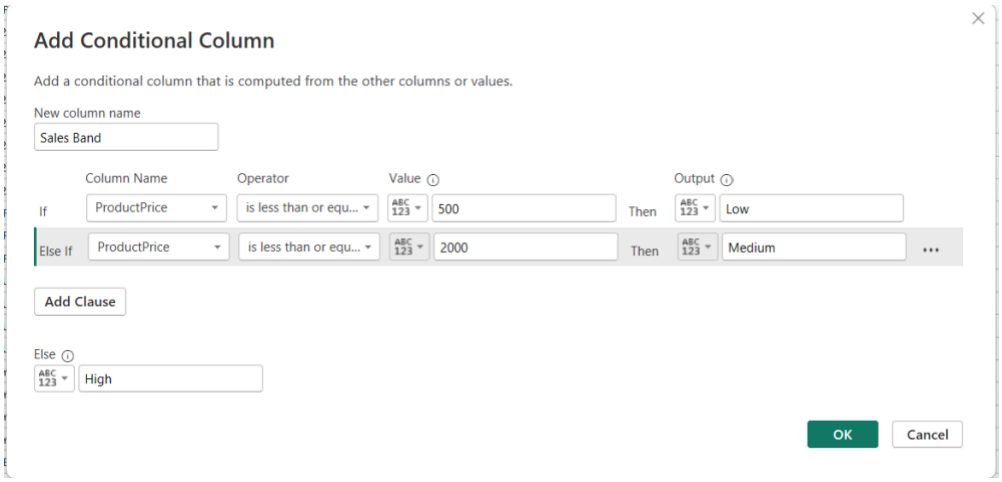
**Steps:** Add Column → Conditional Column on the Sales field:

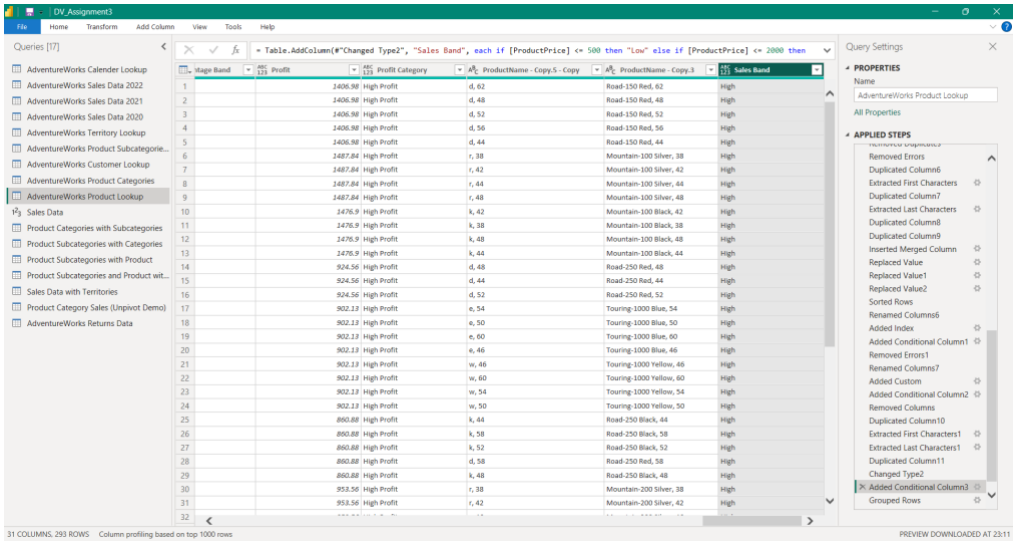
₹0–₹500 → Low

₹501–₹2000 → Medium

Above ₹2000 → High

**Purpose:** Groups every transaction into a Low/Medium/High sales band.





## 2. Create Shipping Performance (Conditional Column)

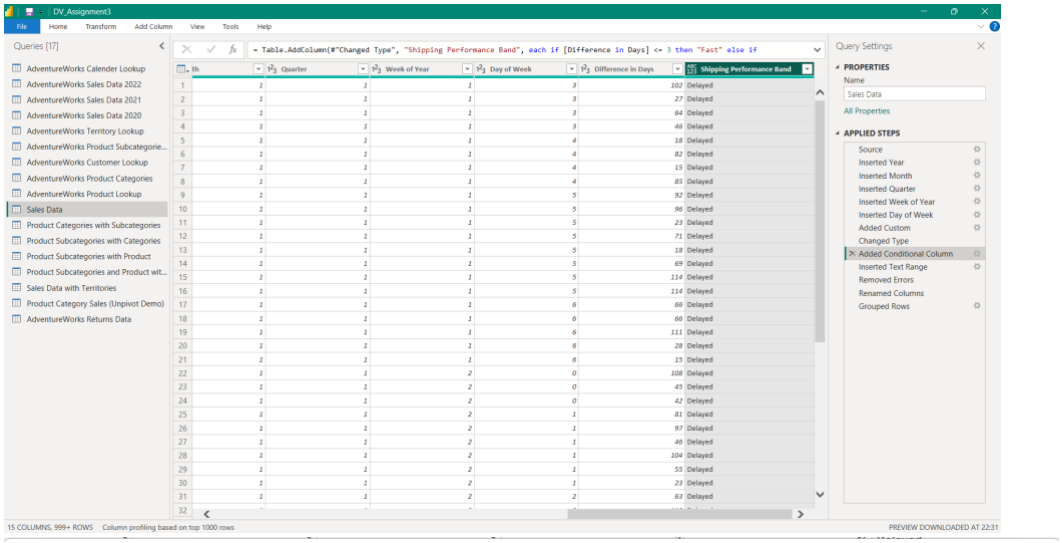
**Steps:** Add Column → Conditional Column on the days-difference (OrderDate − StockDate) field:

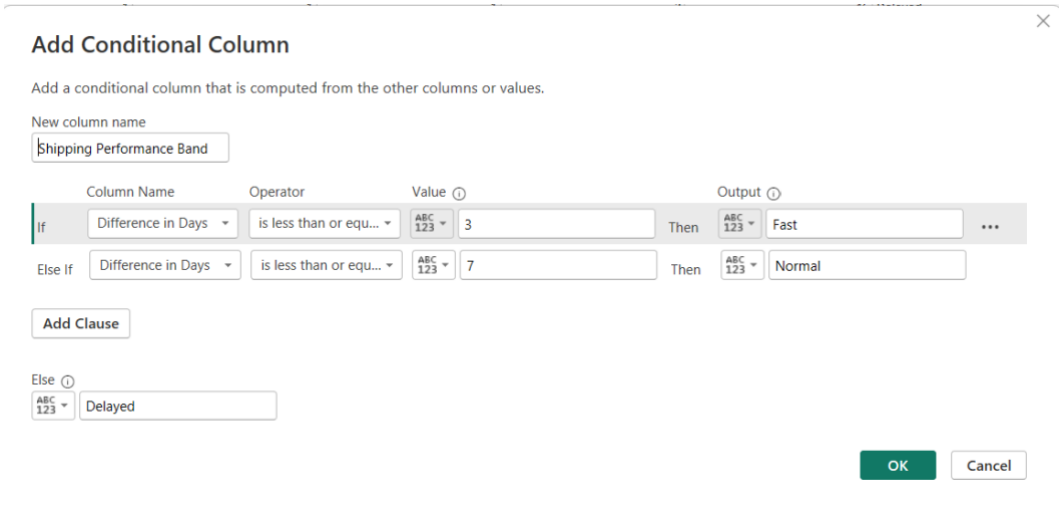
0–3 Days → Fast

4–7 Days → Normal

More than 7 Days → Delayed

**Purpose:** Classifies each order by how quickly it was shipped.



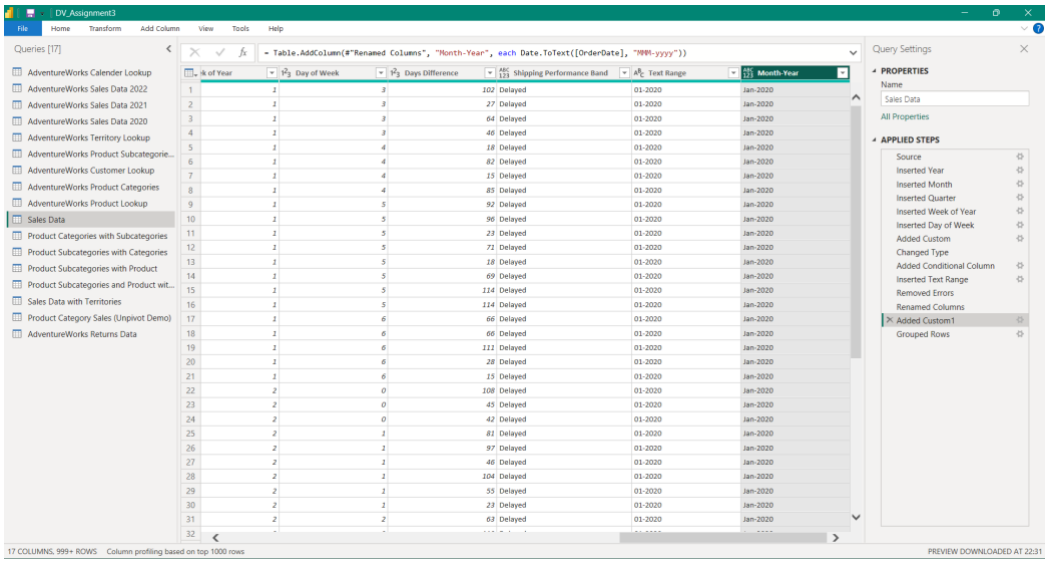


### 3. Extract Month-Year from Order Date

**Steps:** Add Column → Custom Column → combine Date.ToText([OrderDate], "MMM") & "-" & Text.From(Date.Year([OrderDate])).

**Purpose:** Produces a readable Month-Year label such as Jan-2014 for time-series charts.

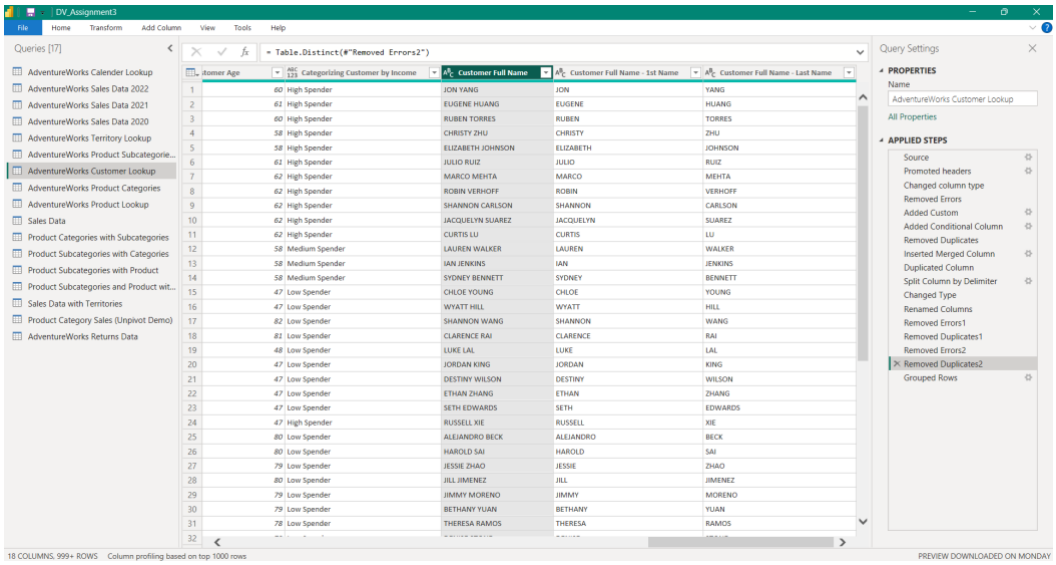


## 4. Split Customer Full Name into First Name and Last Name

**Steps:** Select the Customer Full Name column → Split Column → By Delimiter → Space.

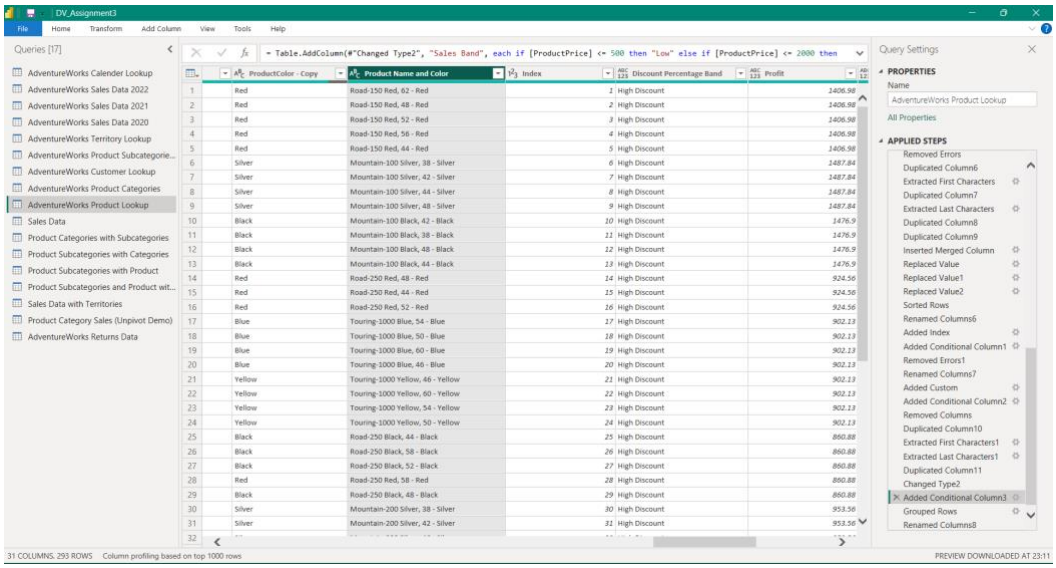
**Purpose:** Separates the full name into individual First Name and Last Name columns.




## 5. Merge Product Name and Color

**Steps:** Ctrl-select Product Name and Color columns → Merge Columns, with a separator such as " - ".

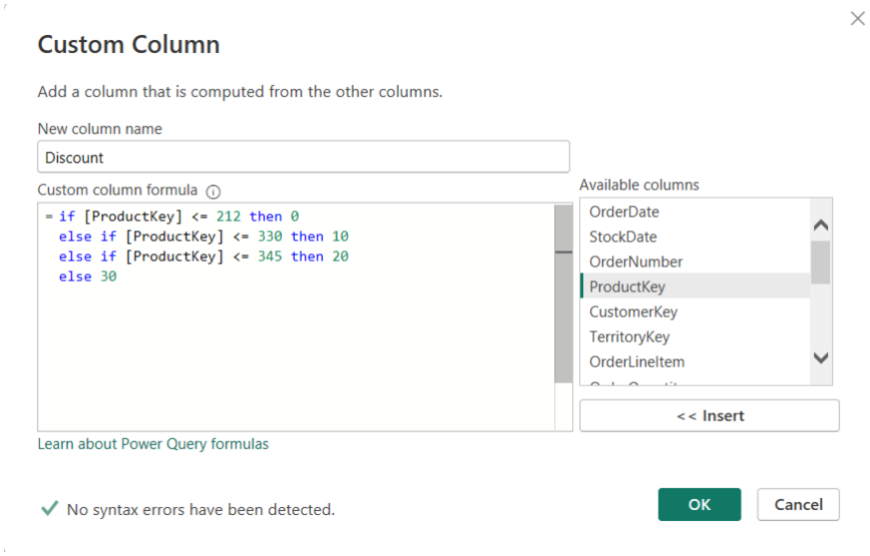
**Purpose:** Creates a single combined name-color label for reporting.

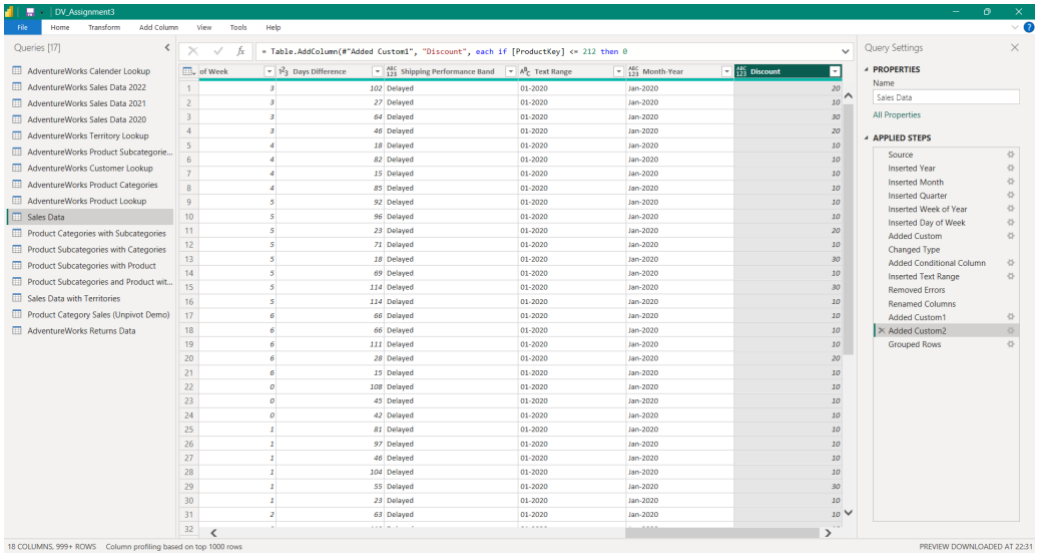


## 6. Create Discount Band

**Steps:** Add Column → Conditional Column on Discount with four tiers: No Discount, Low Discount, Medium Discount, High Discount.

**Purpose:** Buckets orders into discount tiers to analyze discounting patterns.

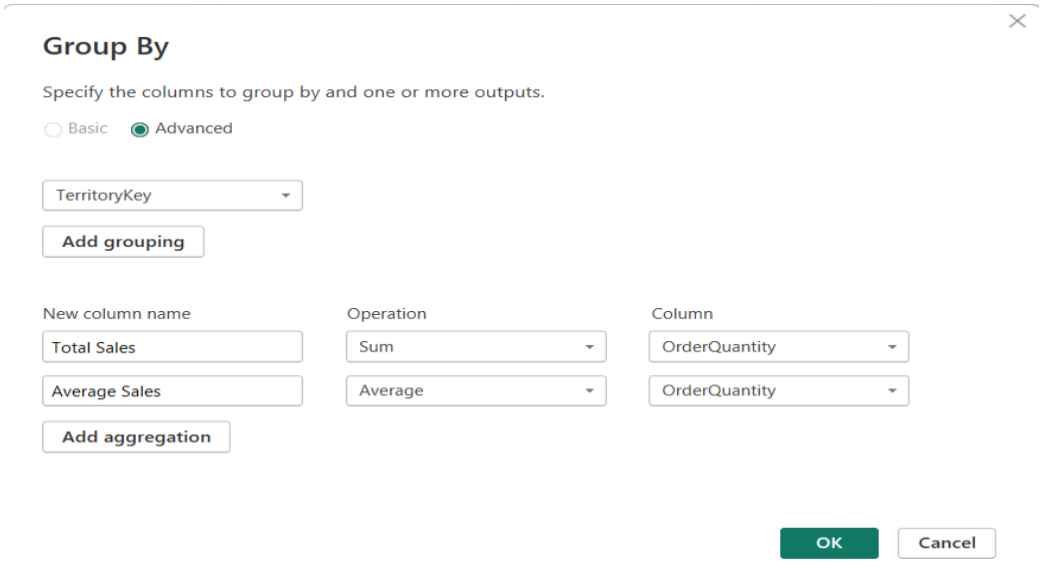


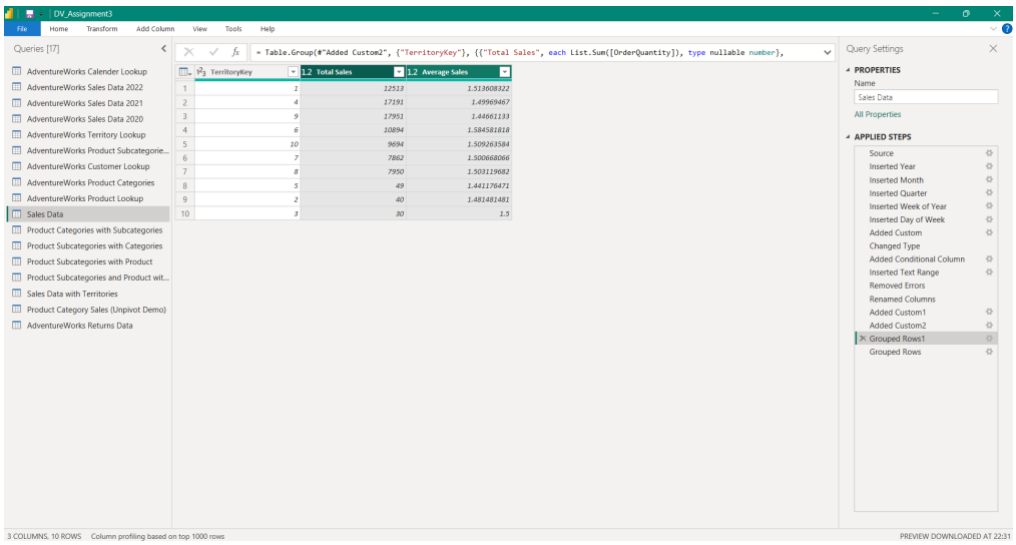



### 7. Group by Region — Total Sales, Average Sales, Total Quantity

**Steps:** Select Region column → Transform → Group By → Add multiple aggregated columns: Sum of Sales, Average of Sales, Sum of Quantity.

**Purpose:** Summarizes total sales, average sales, and total quantity for every region.



### Conclusion

Through this practical, hands-on experience was gained in performing built-in aggregations (SUM, AVERAGE, COUNT, DISTINCTCOUNT, MIN, MAX) and creating a wide range of calculated columns using Power Query — including date/time extraction, conditional categorization, text transformations, column merging/splitting, duplicate handling, sorting, and grouped aggregations. These techniques form the foundation for preparing clean, analysis-ready data before building visualizations and dashboards in Power BI.

**LINKS OF BLOG:**

| Platform | Link |
|---|---|
| Google Docs | https://docs.google.com/document/d/1qn_YvyMdh3DypDJVbZDDmkGAmC06Wcux-wk2iW6lQgE/edit?usp=sharing |
| GitHub | https://github.com/mst-2005/DataVisualization/tree/main/Experiment3 |
| LinkedIn | https://lnkd.in/p/d7FjXbEB |